

NITESH NANKANI

AI Solutions Engineer · Conversational AI · RAG · Multi-Agent Systems

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Customer-facing AI Solutions Engineer. Most engineers start with the code and ask what to build. I start with the human and ask *why* — why do customers abandon carts? Because people love to bargain. So I built an AI agent that bargains like a shopkeeper. Twelve years running a B2B business taught me to find the real problem; the last year taught me to solve it with conversational AI, RAG, and multi-agent systems — five live in production.

EXPERIENCE

AI Solutions Engineer — Independent Sep 2024 – Present

Pune, Maharashtra

Architecting and deploying production AI systems — conversational agents (voice + chat), RAG pipelines, multi-agent intelligence, and fine-tuned BERT classifiers — shipped end-to-end across wellness, e-commerce negotiation, B2B trade intelligence, and customer-complaint analytics, on Railway, Vercel, HuggingFace, and Render. Details of each deployed system listed under AI Projects below.

Core skills: Python · FastAPI · LangChain · ChromaDB · PageIndex · BERT fine-tuning · CrewAI · Claude · GPT-4o · Whisper · Deepgram · ElevenLabs · Twilio · React · Docker · Railway · Vercel · HuggingFace

Teespirit — Performance Marketing & Growth Analyst 2022 – Mid 2024

Direct-to-Consumer Print-on-Demand Apparel (Shopify)

- Managed end-to-end performance marketing for a D2C apparel brand, running 140+ Meta ad campaigns that drove ~3,966 orders across the venture's lifecycle.
- Delivered 24M+ reach, 47M+ impressions, and 160K+ link clicks on ₹29.5L of managed ad spend; tracked CPC, CPM, and add-to-cart funnel metrics daily.
- Owned the full creative-to-conversion funnel: ad creative (mockups, reels, A/B-tested visuals), audience targeting, landing-page optimization, and budget allocation.
- Built Shopify Analytics + Power BI dashboards to monitor campaign-level performance daily; paused underperformers within 48 hours and reallocated spend to top quartile while continuously testing new creative angles.
- Role ended in mid-2024 when the brand restructured operations.

Tools: Meta Ads Manager · Shopify · Shopify Analytics · Power BI · Tableau · Excel · Midjourney · RunwayML · InVideo · Canva (advanced) · Photoshop · CreativeFabrica

Kishor & Company — Business Head 2012 – 2022

Apparel manufacturing (KRYSTAL men's, FOXK kids', CUTEBOY) · Maharashtra, Karnataka, Andhra Pradesh

- Built the wholesale distribution network from 0 agents and a handful of accounts to 38 cities across 3 states — coordinating 4 field agents and covering ~2,000 km of sales territory over the decade.
- Managed 125-order procurement cycles across 8 suppliers, with on-time delivery as the core KPI; identified bottleneck suppliers early using a tracking sheet and pre-routed orders to prevent peak-season stockouts.
- Built a Tableau dashboard on 3 years of SKU-level sales data; identified that plain shirts outperformed printed and shifted production focus toward a high-margin plain-shirts line that became a category we owned in our regional market.
- Used seasonal demand analysis to pre-stock counter-intuitive winners (Royal Blue for May–June, Black for the Dec–Feb winter season) — reducing stockouts in peak demand and freeing working capital in slow months.
- Rationalized a 15-SKU trial range down to 5 always-in-stock core SKUs so wholesale clients could re-order with confidence — reduced churn risk at retail outlets that valued continuity over novelty.

Tools: Tableau · Power BI · Excel (advanced — pivots, forecasting) · WhatsApp Business · Tally

AI PROJECTS

RecallAI — Wellness voice agent (deployed live on Railway)

Twilio · Deepgram · BERT (emotion) · ChromaDB · Claude · ElevenLabs · Sarvam AI

- Built a phone-call-based wellness companion that responds with therapist-style empathy, grounded in a RAG pipeline over 8 psychology and therapy books.

- Architected a 7-step real-time voice pipeline: Twilio call → Deepgram (speech→text) → BERT emotion detection → ChromaDB semantic retrieval → Claude (response generation) → ElevenLabs (text→speech) → playback over the call.
- Added Hindi-language support via Sarvam AI for Indian users, extending the agent to bilingual conversation.
- Diagnosed and fixed an audio-codec mismatch (ElevenLabs ulaw_8000 format → Twilio playback) that was causing distorted voice output.
- Identified sequential-API latency (~1–2 sec per turn) as the primary bottleneck; currently prototyping a migration from Twilio to PipeCat (streaming voice framework) to reduce turn time.

BargainAI — Hinglish negotiation chatbot (deployed live on HuggingFace Spaces · WhatsApp via Twilio)

PageIndex · BERT · ChromaDB · Claude · Twilio · WhatsApp

- Built a conversational AI agent that replaces generic abandoned-cart discount messages with personalized Hinglish bargaining, modeled on how a traditional Indian shopkeeper negotiates.
- Designed a layered retrieval architecture: PageIndex for hierarchical category-aware retrieval (so shirt-negotiation tactics don't mix with electronics) + semantic search within each category — over 10 books on sales, marketing, and negotiation.
- Fine-tuned a BERT intent classifier on 51,000 Flipkart product reviews to detect customer hesitation type (price, quality, comparison, fit) before retrieval, so Claude receives both the customer's intent and the matching negotiation tactic.
- Integrated WhatsApp delivery via Twilio; Shopify integration built and pending live testing.
- Resolved a key engineering challenge: synchronizing BERT (intent detection) with PageIndex (retrieval) inside one RAG pipeline, so the LLM receives both signals together rather than asynchronously.

NiryatAI — Indian export intelligence platform (deployed live · Frontend on Vercel · Backend on Railway)

FastAPI · ChromaDB · Claude · Whisper · Apify · React

- Built an end-to-end AI assistant for Indian exporters that answers natural-language questions on HS codes, government schemes, costing, documentation, and buyer leads — replacing the need to navigate 5+ disconnected government and trade portals.
- Architected a multi-modal RAG pipeline ingesting 114 source files across PDF, DOCX, XLSX, and audio (transcribed via Whisper) — totaling ~20,000 chunks indexed in ChromaDB for semantic retrieval.
- Built a server-side DGFT TradeStat integration to fetch real-time country-wise export data by HS code; since DGFT exposes no official API, reverse-engineered the public form using a GET request (CSRF token extraction) + POST request (form submission with token + HS code + year).
- Integrated Apify (Zauba scraper) for verified international buyer leads across 41 countries, structured by product category and contact details.
- Designed a freemium gating model: AI assistant and trade data free; buyer-lead contact details sit behind a blurred / email-capture paywall — building lead-generation and conversion funnels directly into the product.

Legal RAG Chatbot — deployed on HuggingFace

AI assistant for Indian legal acts (BNS, BNSS, BSA, DPDP Act 2023) using a three-stage architecture: PageIndex (hierarchical retrieval) → BERT (re-ranking) → GPT-4o (answer generation). Stack: FastAPI · Redis · SQLite · Gradio · Docker.

AI Complaint Intelligence Agent — deployed live on HuggingFace Spaces

Multi-agent system that turns unstructured customer reviews into prioritized business intelligence.

- **Complaint Classifier** — assigns each review a complaint type and root cause from 7 product categories.
- **Priority Ranker** — scores and orders complaints by business urgency and customer impact.
- **Trend Detector** — surfaces emerging patterns across review batches before they become escalations.
- **Report Generator** — synthesizes an executive summary: top issues, risk level, and 3 recommended actions.
- **Engine:** 7 fine-tuned BERT classifiers on 51K reviews (99.6–99.8% accuracy) feeding a 4-agent CrewAI pipeline on GPT-3.5.
- **Stack:** FastAPI · Gradio · Docker · HuggingFace.

EDUCATION & CERTIFICATIONS

MBA, Marketing · Pune University 2010

Certifications: Agentic AI Training (Edureka, 2025) · Digital Product Management Specialization (Coursera, 2025) · Google Data Analytics Professional Certificate (Google, 2025)